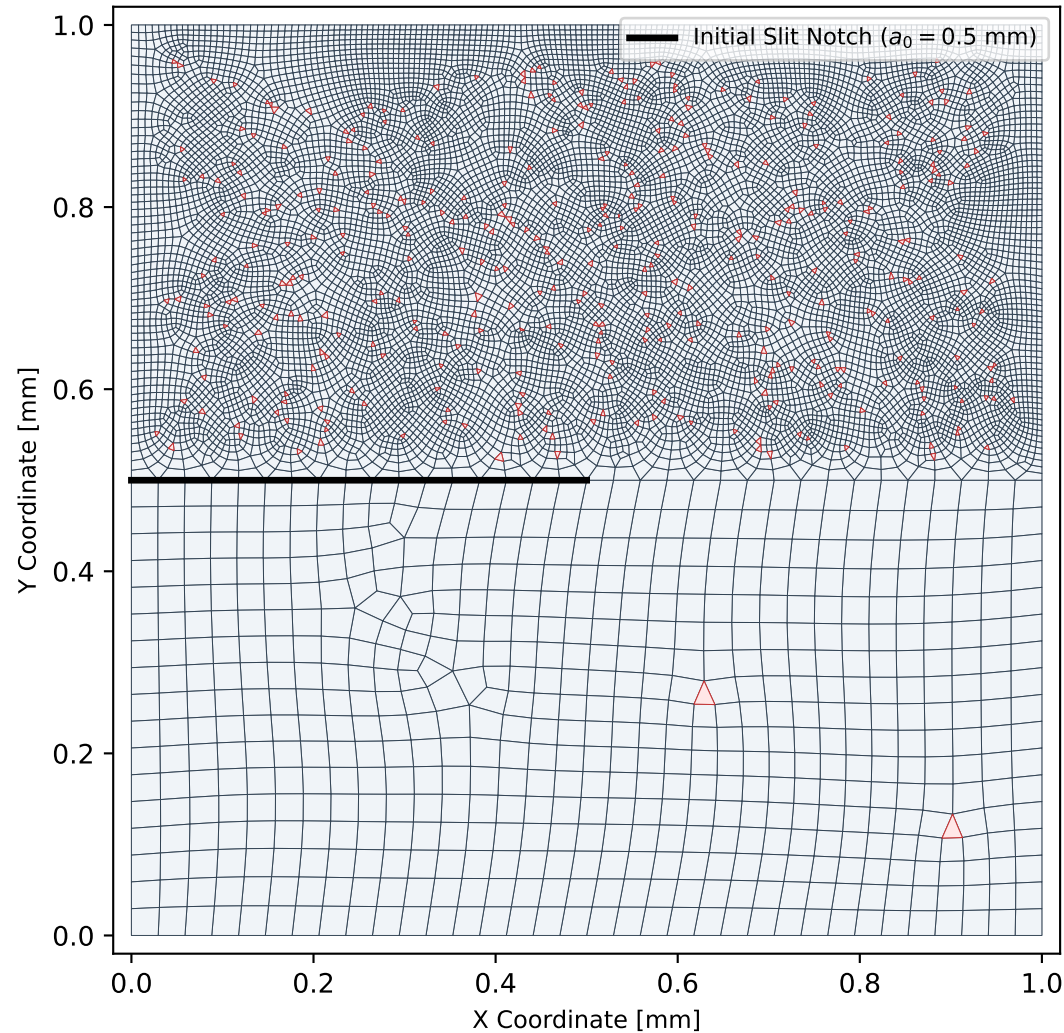


**Pandey & Kumar (2025) Proposed Adaptive Refined Mesh: Topology & Discretization**  
**Model PK\_MODE1\_PROPOSED\_PFM (PBS Jobs: 1398460 / 1398461)**

**Global Domain ( $1.0 \times 1.0$  mm, 14,235 elements)**  
**Blue: Quads (13,854), Pink: Triangles (381)**



**Ligament Process Zone Zoom ( $h_{min} = 0.0010$  mm)**  
**Red Boundary: Quad-to-Triangle Transition**

